

Databases

Labwork 2

Task 1

The table violates **First Normal Form (1NF)** because the `seat_numbers` column contains multiple values in one cell. For example, "12A, 12B" contains two seat numbers. In 1NF, every attribute must contain only one atomic value. Therefore, `seat_numbers` must be separated into individual rows or stored in a separate table.

The table also has dependencies that will need to be resolved when converting it to 2NF and 3NF. For example, `booking_id` → `flight_number`, and `flight_number` → `airline_name`, `departure_airport_name`, and `arrival_airport_name`.

Task 2

Step 1: First Normal Form (1NF)

Convert multi-valued attributes (`seat_numbers`) into atomic values by creating individual rows per seat assignment.

Booking_receipt_1NF (`booking_id`, `seat_number`, `passenger_full_name`, `passenger_passport_number`, `flight_number`, `departure_airport_name`, `departure_city`, `arrival_airport_name`, `arrival_city`, `airline_name`, `ticket_price`)

Primary Key: (`booking_id`, `seat_number`)

Step 2: Second Normal Form (2NF)

Remove partial functional dependencies by separating booking-level data from seat/passenger-level data.

Booking_Receipt (`booking_id`, `flight_number`, `departure_airport_name`, `departure_city`, `arrival_airport_name`, `arrival_city`, `airline_name`, `ticket_price`)

Primary Key: `booking_id`

Booking_Seat (`booking_id`*, `seat_number`, `passenger_full_name`, `passenger_passport_number`)

Primary Key: (`booking_id`, `seat_number`)

Foreign Key: `booking_id` REFERENCES `Booking_Receipt(booking_id)`

Step 3: Third Normal Form (3NF)

Remove transitive dependencies by creating individual entity tables for **Passenger**, **Airport**, **Airline**, **Flight**, **Booking**, and **Passenger_Booking**.

1. **Passenger** (passport_number, passenger_full_name)
 - **Primary Key:** passport_number
2. **Airport** (airport_name, city)
 - **Primary Key:** airport_name
3. **Airline** (airline_name)
 - **Primary Key:** airline_name
4. **Flight** (flight_number, departure_airport_name*, arrival_airport_name*, airline_name*)
 - **Primary Key:** flight_number
 - **Foreign Keys:**
 - departure_airport_name REFERENCES Airport(airport_name)
 - arrival_airport_name REFERENCES Airport(airport_name)
 - airline_name REFERENCES Airline(airline_name)
5. **Booking** (booking_id, flight_number*, ticket_price)
 - **Primary Key:** booking_id
 - **Foreign Key:** flight_number REFERENCES Flight(flight_number)
6. **Passenger_Booking** (booking_id*, seat_number, passport_number*)
 - **Primary Keys:** (booking_id, seat_number)
 - **Foreign Keys:**
 - booking_id REFERENCES Booking(booking_id)
 - passport_number REFERENCES Passenger(passport_number)

Task 3

For 1NF:

No new functional dependency was required. The seat_number attribute was changed to seat_number because each attribute must contain a single atomic value.

- For 2NF:
- booking_id -> flight_number
- booking_id -> ticket_price
- passenger_passport_number -> passenger_full_name
- These dependencies represent partial dependencies on the composite primary key (booking_id, passenger_passport_number). Therefore, the booking and passenger information was separated into different tables.
- For 3NF:
- flight_number -> airline_id
- airline_id -> airline_name
- departure_airport_id -> departure_airport_name, departure_city
- arrival_airport_id -> arrival_airport_name, arrival_city
- The dependency flight_number -> airline_id -> airline_name represents a transitive dependency. Therefore, airline information was separated into the Airline table. Airport

information was also separated into the Airport table, with departure_airport_id and arrival_airport_id stored as foreign keys in the Flight table.

Task 4

The primary key of the Flights table is flight_id.

The main functional dependency is flight_id → all other attributes. Airport information is stored separately in the Airport table, where airport_id determines airport attributes. Airline information is stored separately in the Airline table, where airline_id determines airline attributes. Therefore, there is no transitive dependency between non-key attributes inside the Flights table. The Flights table is in 3NF

DDL 5 — Rename airline_info → airline

lab2/postgres@PostgreSQL 16

Query Query History

```
1 CREATE TABLE airline_info(  
2     airline_id INT NOT NULL,  
3     airline_code VARCHAR(30) NOT NULL,  
4     airline_name VARCHAR(50) NOT NULL,  
5     airline_country VARCHAR(50) NOT NULL,  
6     created_at TIMESTAMP NOT NULL,  
7     updated_at TIMESTAMP NOT NULL,  
8     info VARCHAR(50) NOT NULL,  
9     PRIMARY KEY (airline_id)  
10 );
```

Data Output Messages Notifications

CREATE TABLE

Query returned successfully in 127 msec.

lab2/postgres@PostgreSQL 16

Query Query History

```
1 CREATE TABLE airport (  
2     airport_id INT NOT NULL,  
3     airport_name VARCHAR(50) NOT NULL,  
4     country VARCHAR(50) NOT NULL,  
5     state VARCHAR(50) NOT NULL,  
6     city VARCHAR(50) NOT NULL,  
7     created_at TIMESTAMP NOT NULL,  
8     updated_at TIMESTAMP NOT NULL,  
9     PRIMARY KEY (airport_id)  
10 );
```

Data Output Messages Notifications

CREATE TABLE

Query returned successfully in 48 msec.

lab2/postgres@PostgreSQL 16

Query Query History

```
1 CREATE TABLE passengers (  
2     passenger_id INT NOT NULL,  
3     first_name VARCHAR(50) NOT NULL,  
4     last_name VARCHAR(50) NOT NULL,  
5     date_of_birth DATE NOT NULL,  
6     gender VARCHAR(50) NOT NULL,  
7     country_of_citizenship VARCHAR(50) NOT NULL,  
8     country_of_residence VARCHAR(50) NOT NULL,  
9     passport_number VARCHAR(20) NOT NULL,  
10    created_at TIMESTAMP NOT NULL,  
11    updated_at TIMESTAMP NOT NULL,  
12    PRIMARY KEY (passenger_id)  
13 );
```

Data Output Messages Notifications

CREATE TABLE

Query returned successfully in 49 msec.

lab2/postgres@PostgreSQL 16

Query Query History

```
1 CREATE TABLE booking (  
2     booking_id INT NOT NULL,  
3     flight_id INT NOT NULL,  
4     passenger_id INT NOT NULL,  
5     booking_platform VARCHAR(50) NOT NULL,  
6     created_at TIMESTAMP NOT NULL,  
7     updated_at TIMESTAMP NOT NULL,  
8     status VARCHAR(50) NOT NULL,  
9     price DECIMAL(7,2) NOT NULL,  
10    PRIMARY KEY (booking_id)  
11 );
```

Data Output Messages Notifications

CREATE TABLE

Query returned successfully in 48 msec.

lab2/postgres@PostgreSQL 16

No limit

E

Query Query History

```
1 CREATE TABLE baggage_check (  
2     baggage_check_id INT NOT NULL,  
3     check_result VARCHAR(50) NOT NULL,  
4     created_at TIMESTAMP NOT NULL,  
5     updated_at TIMESTAMP NOT NULL,  
6     booking_id INT NOT NULL,  
7     passenger_id INT NOT NULL,  
8     PRIMARY KEY (baggage_check_id)  
9 );
```

Data Output Messages Notifications

CREATE TABLE

Query returned successfully in 52 msec.

lab2/postgres@PostgreSQL 16

No limit

E

Query Query History

```
1 CREATE TABLE baggage (  
2     baggage_id INT NOT NULL,  
3     weight_in_kg DECIMAL(4,2) NOT NULL,  
4     created_at TIMESTAMP NOT NULL,  
5     updated_at TIMESTAMP NOT NULL,  
6     booking_id INT NOT NULL,  
7     PRIMARY KEY (baggage_id)  
8 );
```

Data Output Messages Notifications

CREATE TABLE

Query returned successfully in 60 msec.

lab2/postgres@PostgreSQL 16

Query Query History

Execute script
F5

```
1 CREATE TABLE boarding_pass (  
2     boarding_pass_id INT NOT NULL,  
3     booking_id INT NOT NULL,  
4     seat VARCHAR(50) NOT NULL,  
5     boarding_time TIMESTAMP NOT NULL,  
6     created_at TIMESTAMP NOT NULL,  
7     updated_at TIMESTAMP NOT NULL,  
8     PRIMARY KEY (boarding_pass_id)  
9 );
```

Data Output Messages Notifications

CREATE TABLE

Query returned successfully in 55 msec.

lab2/postgres@PostgreSQL 16

Query Query History

```
1 CREATE TABLE booking_flight (  
2     booking_flight_id INT NOT NULL,  
3     booking_id INT NOT NULL,  
4     flight_id INT NOT NULL,  
5     created_at TIMESTAMP NOT NULL,  
6     updated_at TIMESTAMP NOT NULL,  
7     PRIMARY KEY (booking_flight_id)  
8 );
```

Data Output Messages Notifications

CREATE TABLE

Query returned successfully in 52 msec.

lab2/postgres@PostgreSQL 16

No limit

Query Query History

1

2

3

4

5

6

7

8

```
CREATE TABLE security_check (  
    security_check_id INT NOT NULL,  
    check_result VARCHAR(20) NOT NULL,  
    created_at TIMESTAMP NOT NULL,  
    updated_at TIMESTAMP NOT NULL,  
    passenger_id INT NOT NULL,  
    PRIMARY KEY (security_check_id)  
);
```

Data Output Messages Notifications

CREATE TABLE

Query returned successfully in 52 msec.

lab2/postgres@PostgreSQL 16

Query Query History

```
1 ALTER TABLE airline_info
2 RENAME TO airline;
3
4 SELECT * FROM airline
```

Data Output Messages Notifications

airline_id	airline_code	airline_name	airline_country	created_at	updated_at
[PK] integer	character varying (30)	character varying (50)	character varying (50)	timestamp without time zone	timestamp without

DDL 6 — price → ticket_price

lab2/postgres@PostgreSQL 16

Query Query History Scratch Pad x

```
1 ALTER TABLE booking
2 RENAME COLUMN price TO ticket_price;
3
4 SELECT * FROM booking
```

Data Output Messages Notifications

booking_id	flight_id	passenger_id	booking_platform	created_at	updated_at	status	ticket_price
[PK] integer	integer	integer	character varying (50)	timestamp without time zone	timestamp without time zone	character varying (50)	numeric (7,2)

DDL 7 — departing_gate VARCHAR → TEXT

The screenshot shows a PostgreSQL IDE window titled 'lab2/postgres@PostgreSQL 16'. The 'Query' tab is active, displaying the following SQL code:

```
1 ALTER TABLE flights
2 ALTER COLUMN departing_gate TYPE TEXT;
3
4 SELECT * FROM flights
```

The 'Data Output' tab is also visible, showing the schema of the 'flights' table:

departing_airport_id	arriving_airport_id	departing_gate	arriving_gate	airline_id	act_departu
integer	integer	text	character varying (50)	integer	timestamp

DDL 8 — Drop info

The screenshot shows a PostgreSQL IDE window titled 'lab2/postgres@PostgreSQL 16'. The 'Query' tab is active, displaying the following SQL code:

```
1 ALTER TABLE airline
2 DROP COLUMN info;
3
4 SELECT * FROM airline
```

The 'Data Output' tab is also visible, showing the schema of the 'airline' table:

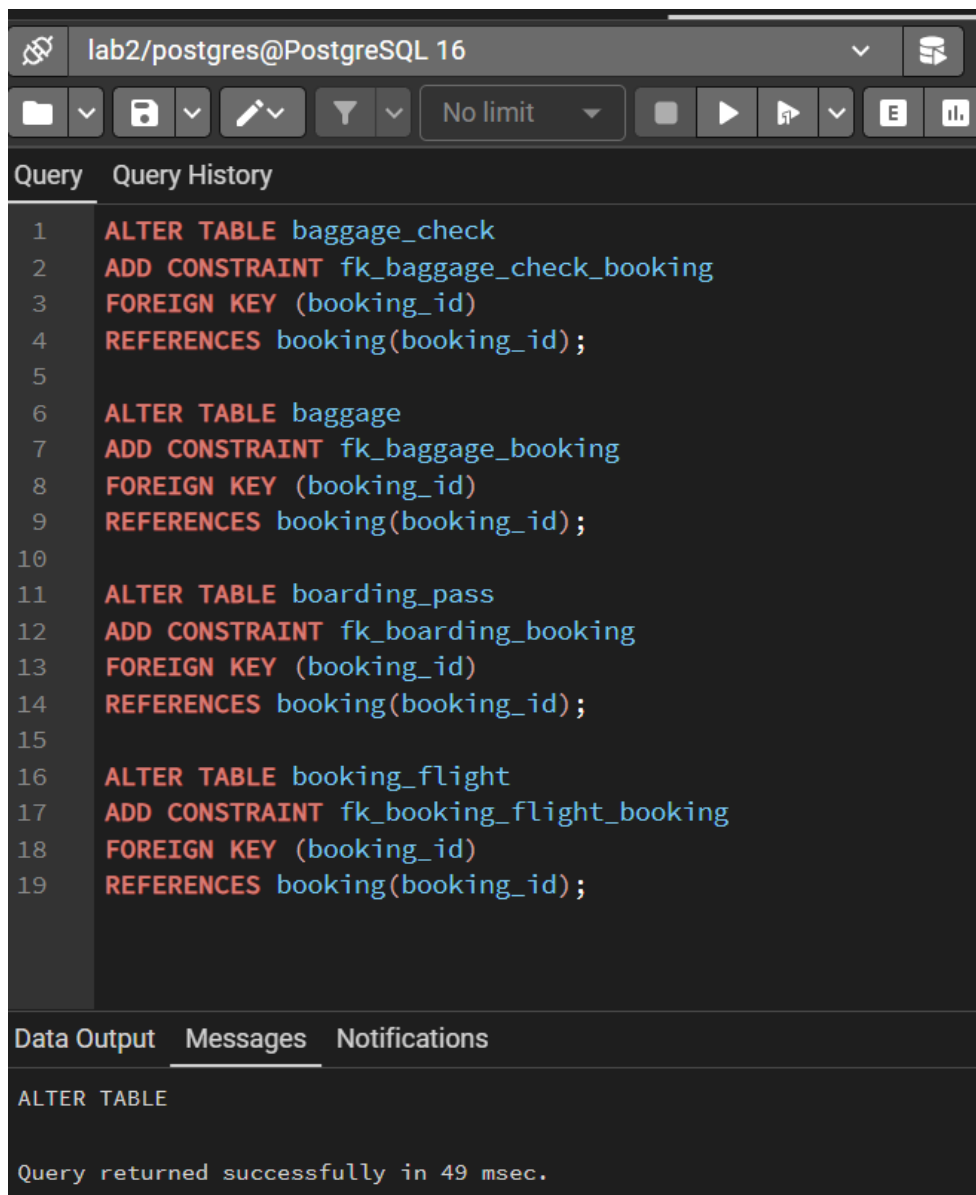
airline_id	airline_code	airline_name	airline_country	created_at	updated_at
[PK] integer	character varying (30)	character varying (50)	character varying (50)	timestamp without time zone	timestamp without time zone

Passengers with Secuitiry_check, Booking, Baggage_check by passenger_id;

The screenshot shows a PostgreSQL IDE window titled 'lab2/postgres@PostgreSQL 16'. The 'Query' tab is active, displaying the following SQL code:

```
1 ALTER TABLE booking
2 ADD CONSTRAINT fk_booking_passenger
3 FOREIGN KEY (passenger_id)
4 REFERENCES passengers(passenger_id);
5
6 ALTER TABLE baggage_check
7 ADD CONSTRAINT fk_baggage_check_passenger
8 FOREIGN KEY (passenger_id)
9 REFERENCES passengers(passenger_id);
10
11 ALTER TABLE security_check
12 ADD CONSTRAINT fk_security_passenger
13 FOREIGN KEY (passenger_id)
14 REFERENCES passengers(passenger_id);
```

Booking with Baggage_check, Baggage, Boarding_pass, Booking_flight by booking_id;



```
lab2/postgres@PostgreSQL 16

Query History

1  ALTER TABLE baggage_check
2  ADD CONSTRAINT fk_baggage_check_booking
3  FOREIGN KEY (booking_id)
4  REFERENCES booking(booking_id);
5
6  ALTER TABLE baggage
7  ADD CONSTRAINT fk_baggage_booking
8  FOREIGN KEY (booking_id)
9  REFERENCES booking(booking_id);
10
11 ALTER TABLE boarding_pass
12 ADD CONSTRAINT fk_boarding_booking
13 FOREIGN KEY (booking_id)
14 REFERENCES booking(booking_id);
15
16 ALTER TABLE booking_flight
17 ADD CONSTRAINT fk_booking_flight_booking
18 FOREIGN KEY (booking_id)
19 REFERENCES booking(booking_id);

Data Output  Messages  Notifications

ALTER TABLE

Query returned successfully in 49 msec.
```

- Flights with Booking_flight by flight_id;
- Airport with Flights by departing_airport_id;
- Airport with Flights by arriving_airport_id;
- Airline with Flights by airline_id;

lab2/postgres@PostgreSQL 16

Query Query History

```
1 ALTER TABLE booking_flight
2 ADD CONSTRAINT fk_booking_flight_flight
3 FOREIGN KEY (flight_id)
4 REFERENCES flights(flight_id);
5
6 ALTER TABLE flights
7 ADD CONSTRAINT fk_flight_departing_airport
8 FOREIGN KEY (departing_airport_id)
9 REFERENCES airport(airport_id);
10
11 ALTER TABLE flights
12 ADD CONSTRAINT fk_flight_arriving_airport
13 FOREIGN KEY (arriving_airport_id)
14 REFERENCES airport(airport_id);
15
16 ALTER TABLE flights
17 ADD CONSTRAINT fk_flight_airline
18 FOREIGN KEY (airline_id)
19 REFERENCES airline(airline_id);
```

Data Output Messages Notifications

ALTER TABLE

Query returned successfully in 48 msec.

lab2/postgres@PostgreSQL 16

Query Query History Scratch Pad X

```
1 INSERT INTO airline
2 (airline_id, airline_code, airline_name, airline_country, created_at, updated_at)
3 VALUES
4 (1, 'KC', 'Air Astana', 'Kazakhstan', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP),
5 (2, 'DV', 'SCAT Airlines', 'Kazakhstan', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP),
6 (3, 'LH', 'Lufthansa', 'Germany', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP),
7 (4, 'TK', 'Turkish Airlines', 'Turkey', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP),
8 (5, 'EK', 'Emirates', 'UAE', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP);
9
10 SELECT * FROM airline
```

Data Output Messages Notifications

Showing rows: 1 to 5 Page No: 1

	airline_id [PK] integer	airline_code character varying (30)	airline_name character varying (50)	airline_country character varying (50)	created_at timestamp without time zone	updated_at timestamp without time zone
1	1	KC	Air Astana	Kazakhstan	2026-09-21 20:25:40.730574	2026-09-21 20:25:40.730574
2	2	DV	SCAT Airlines	Kazakhstan	2026-09-21 20:25:40.730574	2026-09-21 20:25:40.730574
3	3	LH	Lufthansa	Germany	2026-09-21 20:25:40.730574	2026-09-21 20:25:40.730574
4	4	TK	Turkish Airlines	Turkey	2026-09-21 20:25:40.730574	2026-09-21 20:25:40.730574
5	5	EK	Emirates	UAE	2026-09-21 20:25:40.730574	2026-09-21 20:25:40.730574

